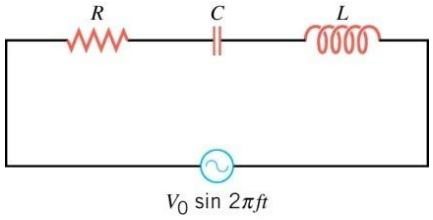
Physics 2 Problem Solving – Chapter 52 RCL circuits

1) A RCL series circuit has a 800 μF capacitor, a 3 mH inductor and a 4 Ω resistor in series with a 10 V rms ac power supply.



A) Determine the impedance of the circuit at 200 Hz.

Answer: 4.87 Ω

B) Determine the current in the circuit if the ac supply has a frequency of 200 Hz.

Answer: 2.05 A

C) Determine the voltage across the resistor at 200 Hz.

Answer: 8.22 A

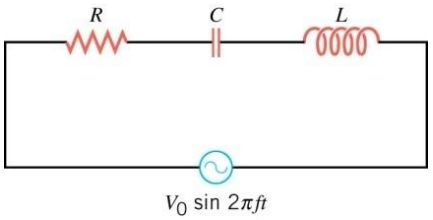
D) Determine the phase angle between the voltage and current in the circuit at 200 Hz.

Answer: 35°

E) Determine the power dissipated by the circuit at 200 Hz..

Answer: P = 16.85 W

2) A RCL series circuit has a 800 μF capacitor, a 3 mH inductor and a 4 Ω resistor in series with a 10 V rms ac power supply.



A) Determine the resonance frequency of the circuit.

Answer: 102.7 Hz

B) Determine the impedance of the circuit at resonance.

Answer: 4 Ω

C) Determine the current in the circuit at the resonance frequency.

Answer: 2.5 A

D) Determine the voltage across the resistor at resonance.

Answer: 10 V

E) Determine the phase angle between the voltage and current in the circuit at resonance.

Answer: 0°

F) Determine the power dissipated by the circuit at resonance.

Answer: 25 W